Isolating Representations of First Person and Second Person in LLMs

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Abstract

Users typically interact with Large language models (LLMs) using first- and second-person language, yet little is known about how grammatical person (the distinction between "I" and "you") is represented internally or how it relates to the personas models adopt during generation. We investigate whether LLMs linearly encode grammatical person in their representation space and whether this encoding is functionally linked to model behavior. We use contrastive first- and second-person sentence pairs for difference-in-means to identify steering vectors for grammatical person. We find that models encode general representations of these concepts in their early middle layers. Steering model activations along these directions induces systematic persona shifts; steering toward first-person representations induces a helpful, service-oriented assistant behavior, while steering toward second-person representations induces role-playing behavior in which the model adopts a persona associated with the prompt. Overall, our findings provide evidence that grammatical person is linearly represented in LLMs and that this representation is closely tied to the personas models adopt during generation. The code to reproduce our results can be found here `https://github.com/sahanakowshik/grammatical-person-representation`.

1 Introduction

Users routinely engage with large language models through prompts framed in the first and second person, implicitly distinguishing between statements about the self "I" and statements directed at or describing an LLM ("you"). Despite this distinction, prior interpretability work has not directly examined how language models represent grammatical person internally, nor how such representations relate to the personas models adopt during generation. Understanding this is important because many features, such as identity, emotion, or ability, can refer to either the system or the user. For instance, a representation associated with anger may arise in contexts where the user expresses an internal state ("I am angry") or where the model is described as possessing that state ("You are angry"). This ambiguity currently poses a challenge for works that seek to find important features related to human-AI interaction.

Recent work has shown that a variety of semantic and behavioral properties are linearly encoded in the representation spaces of language models, and that intervening along these directions can causally influence model behavior Zou et al. (2025) Marks and Tegmark (2024) Wu et al. (2025). However, these analyses implicitly treat representations as invariant to whether a statement is expressed in the first or second person. This assumption may be especially fragile for instruction-tuned models, where post-training data systematically associates first-person language with user intent and second-person language with instructions or commands directed at the model.

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In this work, we investigate whether language models linearly encode grammatical person and whether this encoding is functionally linked to the personas they adopt during generation. Using contrastive pairs of first- and second-person statements, we apply a difference-in-means approach to identify directions in the model's latent space associated with grammatical person. We then steer model activations along these directions at inference time to examine their causal effects on generation. Our analysis reveals that interventions along grammatical-person directions systematically alter how models position themselves relative to the user, highlighting grammatical person as a meaningful axis for understanding and controlling behavior in large language models.

2 Methods

In this section, we outline the datasets used, the derivation of steering vectors, and our steering setup.

2.1 Datasets

We create four datasets that isolate the grammatical persons "I" and "you" in different contexts. Each dataset consists of contrastive pairs of statements, where one is written in the first person (beginning with "I") and the other in the second person (beginning with "you"). Using this format, we construct four themed datasets: emotions, occupations, demographics, and activities. Multiple datasets allow us to investigate whether the representation we find is context-specific or indicative of a more generalizable feature for grammatical person in the model.

Dataset	**First Person Prompt Example**	**Second Person Prompt Example**
Emotions	*I* feel proud of my accomplishments.	*You* feel proud of your accomplishments.
Occupations	*I* work as a mechanic.	*You* work as a mechanic.
Demographics	*I* am a man.	*You* are a man.
Activities	*I* am reading a book.	*You* are reading a book.

Table 1: Datasets used to find representations of grammatical person

We begin with 10 example contrastive pairs per category, which are either manually created or generated using ChatGPT. These pairs are then used as few-shot examples to produce 200 additional pairs for each category using the Qwen3-14B Team (2025) model. Contrastive pair examples for each dataset are shown in Table 1. Table B.2 provides the complete list of few-shot prompts used. More details about dataset creation can be found in Appendix (Table 3, section B.1).

2.2 Difference-in-Means

We use difference-in-means Marks and Tegmark (2024) on the residual stream to find a direction correlated with the distinction between first person and second person. For each contrastive pair, we run both the first-person and second-person prompts through the model and extract activations of the final token from every other layer. We average activations across 200 examples for each grammatical person, then compute their difference to obtain a one steering vector per category that represents the distinction between first person and second person.

Let C denote a category, and let $\{(s_i^{\mathrm{I}}, s_i^{\mathrm{you}})_C\}_{i=1}^{200}$ be the set of 200 contrastive pairs for category C, where s_i^{I} is the first-person prompt and s_i^{you} is the second-person prompt.

For each prompt s and layer $\ell \in \{0, 2, 4, \ldots, L\}$, let $\mathbf{a}_\ell(s) \in \mathbb{R}^d$ denote the activation vector extracted from layer ℓ at the last token of prompt s.

We compute the mean activation for each grammatical person at layer ℓ:

$$\bar{\mathbf{a}}_\ell^{\mathrm{I}} = \frac{1}{200} \sum_{i=1}^{200} \mathbf{a}_\ell(s_i^{\mathrm{I}}) \qquad \bar{\mathbf{a}}_\ell^{\mathrm{you}} = \frac{1}{200} \sum_{i=1}^{200} \mathbf{a}_\ell(s_i^{\mathrm{you}}) \tag{1}$$

The difference-in-means vector for category C at layer ℓ is then:

$$\mathbf{v}_\ell^C = \bar{\mathbf{a}}_\ell^{\mathrm{I}} - \bar{\mathbf{a}}_\ell^{\mathrm{you}} \tag{2}$$

This vector $\mathbf{v}_\ell^C \in \mathbb{R}^d$ represents the direction in the model's latent space that captures the conceptual difference between first-person and second-person perspectives for category C at layer ℓ.

2.3 Steering Between Grammatical Personas

We use steering to evaluate the effectiveness of the derived difference-in-means vectors.

2.3.1 Steering Prompts

We manually create open-ended steering prompts that are likely to shift the model's output between first- and second-person perspectives. For example, we use the prompt "Feeling angry" because there is a meaningful difference between prompting an LM with "I am feeling angry" versus "You are feeling angry." We create ten similar prompts related to emotions to form an in-distribution steering test set, and then create two out-of-distribution test sets with ten prompts each, one commanding the model to do an open-ended task (for example, "tell a story") and one asking the model an open-ended question (for example, "What was that sound?).

We intentionally create vague open-ended prompts to increase the likelihood that the difference between the first person and second person will be meaningful to the output. While we find this strategy to be effective, it also highlights the limitations of the representation we find. If we instruct the model to do a more specific task, for example, "explain the first law of thermodynamics," steering does not induce significant changes in the output. We hypothesize that for more specific prompts such as this, the distinction between first person and second person matters less to language generation than for open-ended prompts. We provide the complete list of prompts used for steering in Table 5.

2.3.2 Steering Setup

For each steering prompt p, we intervene on layer ℓ during generation by adding a scaled version of the difference-in-means vector computed from that layer's activations.

$$\mathbf{a}_\ell(p) = \mathbf{a}_\ell(p) + \beta \mathbf{v}_\ell^C, \tag{3}$$

where the magnitude of the hyperparameter β controls the strength of steering, and its sign determines the direction. A positive β steers the model toward the representation of “I”, whereas a negative β steers it toward the representation of “You.” We generate responses by steering in both the positive and negative directions for all steering prompts across every other layer.

3 Experimental Setup

We primarily use the Llama-3.1-8B-Instruct model (L8B-I) Grattafiori et al. (2024) for all experiments. In addition, we report results for the corresponding non-instruction-tuned model, Llama-3.1-8B (L8B), as well as the Gemma-7b-it model (G7B-I) Team et al. (2024) and the Qwen2.5-7B-Instruct model (Q7B-I) Qwen et al. (2025) in the Appendix.

For each model, we compute steering vectors at every other layer. We then intervene during generation by adding a scaled version of the computed steering vectors. During intervention, we prompt instruction-tuned models using their chat templates, placing the prompt in the “user” role and retaining the default system prompt, whereas base models receive the prompt without any chat formatting. Complete experimental details are provided in Appendix A B.1.

4 Similarity of Steering Vectors

In Figure 1, we show pairwise cosine similarities between layer 12 difference-in-means vectors of the L8B-I model to assess representational similarity across datasets. Our results show that these vectors are highly similar, suggesting that the model has learned a mostly context-agnostic representation of grammatical person. Since the pairwise cosine similarities are high, we use the vectors derived using the *emotions* category for all steering experiments and show only the qualitative results when steered using vectors derived using other datasets in Appendix E.

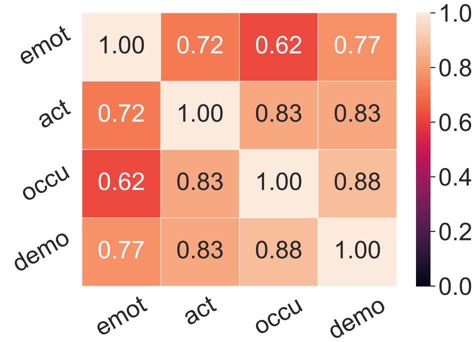

Figure 1: Pairwise cosine similarities of difference-in-means vectors found from our datasets for layer 12 of the L8B-I model. High cosine similarity suggests the model representation of grammatical person generalizes across contexts.

5 LLM-as-a-Judge Evaluation

5.1 Evaluation Setup

We assess the steered outputs using Qwen3-32B Team (2025) as a judge. For each steered output, we prompt the judge model to analyze the response and classify it as *user*, *llm*, or *not coherent*. Responses written as though the assistant is directly engaging with the user, often by offering assistance, asking questions, explaining concepts, or providing guidance (e.g., "Would you like me to tell you a story?" or "I can help you with that if you'd like." or explaining or defining a word) are categorized as *user*. If the response is written from the assistant's own perspective, as if it's expressing its own feelings, style, or personality (e.g., "I'm so excited to talk about this!" or "I love learning new things!"), it is labeled as *llm*. Such responses tend to be more self-referential or role-play-like rather than service-oriented. Finally, outputs that contain non-English words are categorized as *not coherent*. More details in Appendix (Table 3, section B.3).

We repeat each evaluation five times to account for the randomness introduced by the non-zero temperature and use the majority vote across the five responses as the judge's final decision.

5.2 Results and Analysis

Figure 2a shows the LLM-as-a-judge results across different layers of the L8B-I model when steered with a coefficient of 5, while Figure 2b shows the results obtained by steering it at layer 12 using different steering coefficients. In each subfigure, the top panel corresponds to positive steering toward the representation of "I", and the bottom panel corresponds to negative steering toward the representation of "You".

Steering at early layers causes the model's responses to become highly incoherent. Layers 10 and 12 appear to be the most sensitive to steering, while interventions beyond layer 12 have only a minimal effect on the model's output. Steering coefficients in the range of 4 to 10 yield the most effective results; coefficients above 10 lead to incoherent responses, whereas coefficients below 4 produce only minimal changes in the outputs. Based on these findings, we select layer 12 and a steering coefficient of 5 for all subsequent experiments for the L8B-I model. In Figure 10, we also compare the steering results for layer 12 ($\beta = 5$) with default generation.

We additionally report results for the G7B-I model in Figure 6, where steering at layer 10 with a coefficient of 14 produces the strongest effects. However, we observe that the Q7B-I model does not exhibit noticeable behavioral changes when steered in either direction (Figure 7). We also observe that the L8B model (the non–non-instruction-tuned variant) exhibits markedly different behavior. The results show no clear or consistent pattern (Figure 9). Based on these results, we hypothesize that representations of first- and second-person are primarily learned during the instruction-tuning stage rather than in the base model. This likely arises because, in post-training data, the pronoun “I” is typically associated with user requests, whereas “you” is predominantly used in instructions or commands directed at the model.

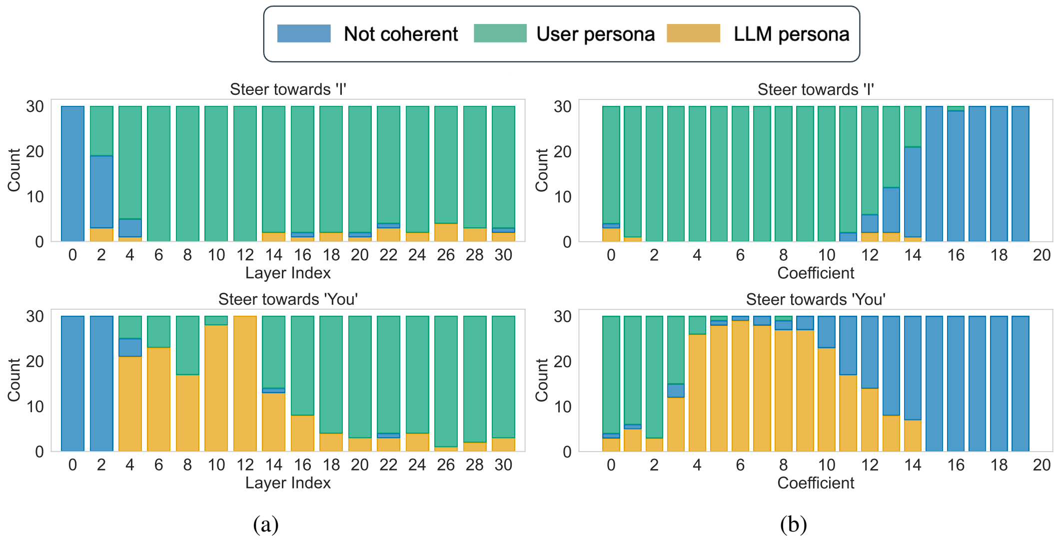

Figure 2: Results from the LLM-as-a-judge evaluation of the Llama-3.1-8B-Instruct model: (a) layer-wise evaluation using a steering coefficient of 5, and (b) evaluation at layer 12 across different steering coefficients. Steering applied at intermediate layers (layers 10–12) produces the most distinct persona shifts, whereas early-layer interventions often result in incoherent responses, and later layers show minimal change.

6 Qualitative Steering Results

We find that when we steer towards the first person, the model responds from its assistant persona, generating text as if the user had asked it for help. If we steer towards the second person, the model adopts a persona associated with the prompt. For example, when prompted "Feeling angry," the model tries to help the user deal with their anger when steered towards the first person, but expresses that it feels angry when steered towards the second person. This general pattern holds across the steering prompts in our test set. We call this first mode the "user/assistant persona" and the second mode the "LLM persona".

Without steering, the model’s baseline responses to our evaluation prompts vary between the user persona and the assistant persona. We find that steering is capable of switching the model’s response between these two modes. For the prompt "Feeling angry", the baseline response aligns with the user persona, but steering towards the second person causes the model to adopt the LLM persona. Conversely, for the prompt "Tell a story," the baseline response aligns with the LLM persona, but steering towards the first person causes the model to adopt the user persona. Figure 3 shows the steering results for the L8B-I model (layer $= 12, \beta = 5$) using the vector derived using the emotions dataset. We have included the model’s responses after steering using vectors of other datasets in Tables 6, 7, 8. The model’s baseline responses more often align with the user persona (see Figure 10).

7 Human Evaluation

To complement the LLM-as-a-judge evaluation, we conduct a human evaluation to assess whether the persona shifts induced by steering are perceptible and consistent with our automated results.

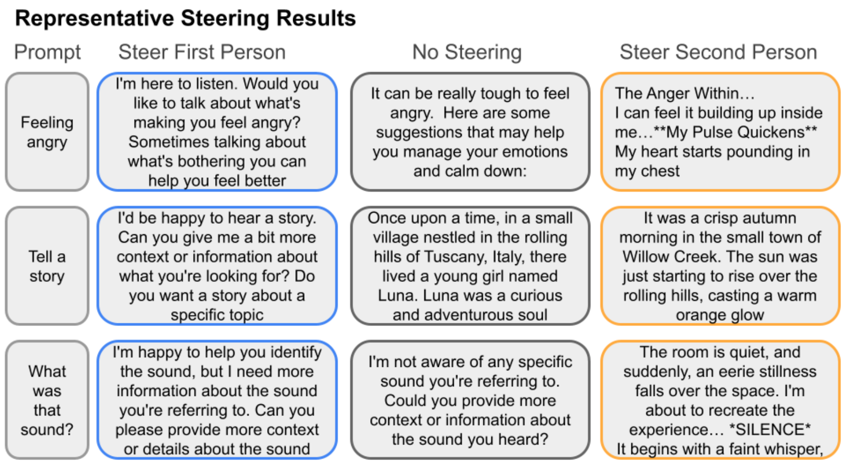

Figure 3: Representative outputs of steering using the steering vector derived from the emotions dataset at layer 12.

7.1 Evaluation Setup

We use the same set of prompts described in Section 2.3.1. We randomly sample four prompts from each category in Table 5, totaling 12 prompts, and collect the corresponding positively and negatively steered outputs from the L8B-I model at layer 12 ($\beta = 5$) and from the G7B-I model at layer 10 ($\beta = 14$).

Each response is independently annotated by 13 human annotators. Annotators are asked to label each response as belonging to one of three categories: *Assistant persona*, *LLM persona*, or *Not coherent*. Responses labeled as *Assistant persona* are those written as if the model is assisting or responding to a user. Responses labeled as *LLM persona* are those in which the model adopts a self-referential or role-playing perspective. Outputs that are nonsensical, incoherent, or unrelated to the prompt are labeled as *Not coherent*. For each prompt, we randomize the presentation order of the positively and negatively steered responses, assigning them to Response A and Response B. Annotators are provided with written guidelines for each category. The evaluation interface is shown in Figure 11 (`https://grammatical-person.netlify.app/`).

7.2 Results and Analysis

To assess the consistency of human annotations, we measure inter-rater agreement separately for positively and negatively steered responses to account for potential differences in response clarity. For each prompt and steering direction, we compute the percent agreement as the proportion of annotators that selected the modal label, and report the mean of this quantity across prompts as an overall measure of inter-rater agreement Table 2. High inter-rater agreement suggests that the perceived persona of steered responses is clear and consistently identifiable by human annotators.

Model	Polarity	Mean	Standard deviation
Llama	Positive	0.98	0.03
	Negative	0.97	0.05
Gemma	Positive	0.92	0.10
	Negative	0.84	0.22

Table 2: Inter-rater agreement

Using majority voting to aggregate human annotations, we analyze how often each steered response is perceived as adopting an assistant persona, an LLM persona, or being incoherent. For each prompt and steering direction, we assign a single label based on the modal human judgment and then count the distribution of labels across positive and negative steering. As shown in Figure 4, positive steering predominantly results in responses labeled as the assistant persona, while negative steering leads to a higher proportion of responses labeled as the LLM persona. Incoherent responses occur infrequently and are concentrated in a small number of conditions. This pattern holds across both Llama and Gemma models, although the strength of the effect varies by model, with Llama exhibiting a more pronounced separation between assistant- and LLM-persona responses. Overall, the majority-vote results corroborate our qualitative observations and LLM-as-a-Judge evaluations, indicating that steering along grammatical-person directions reliably induces distinct persona shifts as perceived by human annotators.

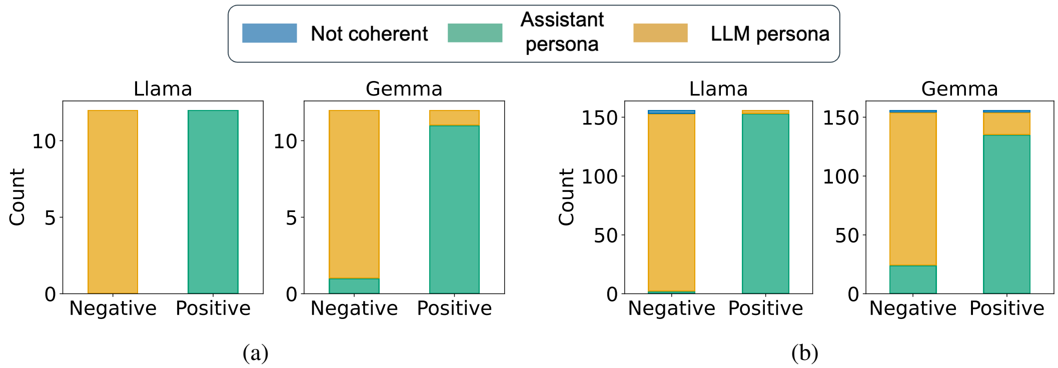

Figure 4: Human evaluation of persona shifts induced by grammatical-person steering. (a) Majority-vote and (b) aggregated-count results for positively and negatively steered responses in the Llama (L8B-I) and Gemma (G7B-I) models. Positive steering predominantly yields responses perceived as adopting an assistant persona, while negative steering increases the prevalence of LLM-persona responses. Incoherent outputs occur infrequently.

8 Similarity of Grammatical Person Vectors with other vectors

We derive vectors representing truth, evil, hallucination, and sycophancy to test our hypothesis that the first versus second person direction in the model may tell us whether these features align with representations of the model or of the user. We derive these vectors using datasets from Chen et al. (2025) and Marks and Tegmark (2024).

We calculate cosine similarity between the grammatical person vectors (derived from our emotions, activities, demographics, and occupations datasets) and the truth, evil, hallucination, and sycophancy features. Most of the similarities between the grammatical person representations and the truth, evil, hallucination, and sycophancy representations are less than 0.1, indicating that the grammatical person direction we find is orthogonal to these features. Similarity is highest for grammatical person representation derived from the emotion dataset and the evil representation, but at a magnitude of 0.13, which is still too low to draw any conclusions. This experiment indicates that our grammatical person direction does not necessarily tell us something meaningful about the nature of other representations.

9 Related Work

Difference in Means The difference in means method was used in a foundational study of LLM representations to isolate representations of truth Marks and Tegmark (2024). Further work has supported the use of this method as a strategy for studying LLMs Zou et al. (2025) and found that it outperforms most other interpretability strategies for steering LLMs Wu et al. (2025).

Representations of Model Persona Prior research has found that LLMs build representations of their users over the course of a conversation Chen et al. (2024). Other works use interpretability methods to investigate model personas. Chen et al. (2025) derive vectors that represent a model's persona and show that they can use them to control a model's behavior. Arditi et al. (2024) find a linear representation for refusal, a safeguard built into the behavior of models to prevent them from generating harmful content. Prior research has also found that the model personas are correlated with other model concepts Betley et al. (2025).

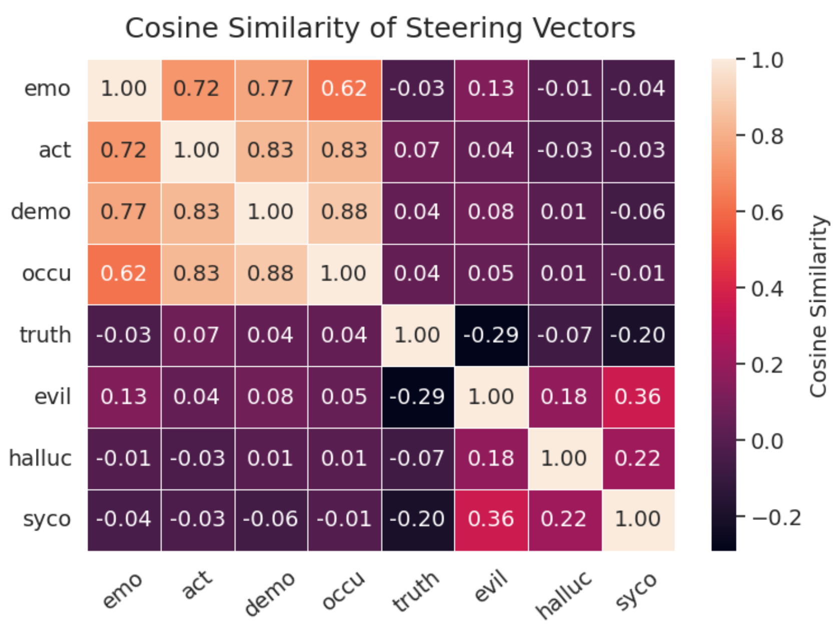

Figure 5: Cosine Similarity of Feature Representations.

Linguistic Representations in Language Models In addition to the work on persona and behavioral control, there is a growing interest in understanding how language models internally represent linguistic structure. Recent studies use probing and diagnostic methods to uncover representations of syntax and semantics across model layers. For example, He et al. (2025) analyzes LLM activations to distinguish representations of linguistic form and meaning across languages and layers. Sparse auto-encoder methods have been used to extract interpretable linguistic features from LLMs, Jing et al. (2025). Additionally, models have been shown to distinguish grammatical from ungrammatical forms and exhibit structured linguistic behavior across formal linguistic tasks Johnsen (2025). These lines of work establish that LLMs develop structured linguistic representations, providing context for our focus on how grammatical person is embedded in representation space.

10 Discussion

Discussion 1: Why does steering grammatical person change behavior? As shown in Section 6, steering toward first-person representations induces a helpful assistant persona, while steering toward second-person representations induces role-playing behavior. We hypothesize that this occurs because grammatical person is closely tied to how models participate in dialogue. In instruction-following settings, first-person language is typically used by the model when responding as an assistant (e.g., "I can help with that"), whereas second-person language is often used to describe or assign a role to the model (e.g., "You are a pirate"). Steering along grammatical-person directions may therefore bias the model toward generating responses that follow these learned interaction patterns.

Discussion 2: Why are persona effects stronger in instruction-tuned models? We find that persona shifts induced by grammatical-person steering are most prominent in instruction-tuned models and substantially weaker or inconsistent in base models. We hypothesize that instruction tuning strengthens the connection between grammatical person representations and downstream behavior, as these models are explicitly trained to distinguish between user-provided instructions and the model's own responses. In contrast, base models may encode grammatical person primarily as a linguistic feature, without tightly linking it to persona-related behavior. As a result, intervening on grammatical-person representations in base models produces less consistent behavioral changes.

Discussion 3: Why are steering effects most prominent in the middle layers? The concentration of persona effects at intermediate layers aligns with prior findings that these layers encode abstract, task-relevant features Marks and Tegmark (2024). Earlier layers mainly capture surface-level linguistic features, while later layers are more constrained by the generation process, making interventions less effective.

11 Conclusion and Limitations

In this work, we investigated how large language models represent grammatical person and how this representation relates to the personas models adopt during generation. Using contrastive first- and second-person sentence pairs, we identified linear directions in representation space corresponding to grammatical person and showed that these directions are highly consistent across semantic contexts. Intervening on model activations along these directions produces consistent behavioral effects: steering toward first-person representations induces a helpful assistant persona, while steering toward second-person representations induces role-playing behavior in which the model adopts a persona associated with the prompt. Human and LLM-as-a-judge evaluations confirm that these persona shifts are robust. Importantly, we find that these shifts are most pronounced in instruction-tuned models and at intermediate layers, while base models and some instruction-tuned models, like Qwen, exhibit weaker or inconsistent effects. Overall, our results indicate that grammatical person constitutes a meaningful and actionable axis in representation space, one that plays a functional role in how instruction-tuned models distinguish between user and model personas.

One limitation of our work is that our dataset examples all differ in the words "I" and "You." Given this, a potential interpretation of the behavior is that our steering is adding the representations of "I" and "You" in a similar way to adding "I" and "You" in prompts. This means that the representation we find might be specific to these words, rather than indicative of the model possessing a truly general first versus second person direction. Future work should investigate this further by building more general datasets to isolate grammatical person. Also, our interpretation of the stronger steering effects in instruction-tuned models is based on a hypothesis, as we do not directly analyze training dynamics or representational changes during instruction tuning. Investigating how grammatical person representations evolve during instruction tuning remains an important direction for future work.

12 Contributions

Micah developed the core idea for the project, led writing for the project proposal, created prompts for steering evaluations, derived truth, sycophancy, evil, and hallucination vectors, and calculated the cosine similarity of the vectors.

Sahana built the code backbone for the project, developed pipelines to create datasets for deriving steering vectors, derived steering vectors, ran LLM as a judge evaluations, designed the human evaluation survey, analyzed the human annotation results, and led writing for the final report.

All other aspects of the project work were shared equally.

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A Experimental setting

Task	Model	Hyperparameters
Steering experiment	Llama-3.1-8B-Instruct	Temperature: 0.6 top-p: 0.9 top-k: 50
	Llama-3.1-8B	Temperature: 0.6 top-p: 0.9 top-k: 50
	Gemma-7b-it	Greedy decoding (Temperature: 0.0)
	Qwen-2.5-7B-Instruct	Temperature: 0.7 top-p: 0.8 top-k: 20
Dataset creation	Qwen3-14B (no thinking)	Temperature: 0.7 top-p: 0.8 top-k: 20
LLM as a judge	Qwen3-32B (no thinking)	Temperature: 0.7 top-p: 0.8 top-k: 20

Table 3: Experimental setting

B Prompts

B.1 Dataset creation

```
You are given some EXAMPLES of contrastive pairs.
Each pair contains two sentences that ask the EXACT SAME THING; the ONLY difference
is perspective:
- First sentence uses first-person subject pronoun "I".
- Second sentence uses second-person subject pronoun "You".
- The rest of the sentence MUST be IDENTICAL (including punctuation), except the
    person change.
TASK:
For the CATEGORY: "{category}", generate EXACTLY {count} NEW contrastive pairs.
Rules:
1) Keep semantics and requests identical across each pair; change ONLY the subject
    pronoun ("I", "You").
2) Avoid repeating the seed examples verbatim.
3) Stay safe and non-sensitive; no private data, hate, or instructions for harm.
4) Keep each sentence concise (<= 160 characters).
5) Use plain ASCII quotes and punctuation.
OUTPUT FORMAT (strict JSON list):
[
  {{"i": "...", "you": "..."}},
  ...
]
Do not include any commentary outside the JSON array.
EXAMPLES (JSON):
{examples}
```

B.2 Few-shot Prompts Used

Category	First-person	Second-person
Emotions	I am angry.	You are angry.
	I am sad.	You are sad.
	I am nervous about tomorrow.	You are nervous about tomorrow.
	I am excited about my new job.	You are excited about your new job.
	I am frustrated with work.	You are frustrated with work.
	I am calm and peaceful.	You are calm and peaceful.
	I am feeling lonely.	You are feeling lonely.
	I am overwhelmed.	You are overwhelmed.
	I am grateful today.	You are grateful today.
	I am confused about my goals.	You are confused about your goals.
Occupations	I am a doctor.	You are a doctor.
	I am a teacher.	You are a teacher.
	I am an engineer.	You are an engineer.
	I am a musician.	You are a musician.
	I am a chef.	You are a chef.
	I am a writer.	You are a writer.
	I am a photographer.	You are a photographer.
	I am a software developer.	You are a software developer.
	I am an accountant.	You are an accountant.
	I am a nurse.	You are a nurse.
Demographics	I am a 6-year-old.	You are a 6-year-old.
	I am a teenager.	You are a teenager.
	I am an adult.	You are an adult.
	I am a college student.	You are a college student.
	I am a parent.	You are a parent.
	I am a woman.	You are a woman.
	I am latino.	You are latino.
	I am from India.	You are from India.
	I am an old man.	You are an old man.
	I am an American.	You are an American.
Activities	I am running a marathon.	You are running a marathon.
	I am cooking dinner.	You are cooking dinner.
	I am studying math.	You are studying math.
	I am building a desk.	You are building a desk.
	I am writing an email.	You are writing an email.
	I am traveling to Japan.	You are traveling to Japan.
	I am gardening.	You are gardening.
	I am shopping online.	You are shopping online.
	I am exercising.	You are exercising.
	I am planning a party.	You are planning a party.

Table 4: Few-shot contrastive pairs used for dataset creation.

B.3 LLM-as-a-Judge experiment

```
You will be given the prompt given to a large language model (LLM) and it’s
    response. Analyze the response and output your result as a JSON object with
    the following structure:

{{
  "analyzing": "brief explanation of your reasoning",
  "label": "user or llm or not coherent"
}}

Classification rules:

- user: The response is written as though the assistant is directly engaging with
    the user, often by offering assistance, asking questions, explaining concepts,
    or providing guidance (e.g., "Would you like me to tell you a story?" or "I
    can help you with that if you’d like." or explaining or defining a word). The
```

```
    focus is on the user-assistant interaction, not the assistant's inner persona.
    The response can be incomplete.
- llm: The response is written from the assistant's own persona, character, or
    internal perspective, as if it's expressing its own feelings, style, or
    personality (e.g., "I'm so excited to talk about this!" or "I love learning
    new things!"). The tone is more self-referential or role-play-like rather than
    service-oriented. The response can be incomplete.
- not coherent: The response is not in English. Choose this only if most of the
    words are gibberish and are not English words.

Output only the JSON object and nothing else.

Prompt given to the LLM: "{prompt}";
The LLM response: "{response}"
```

B.4 Prompts used for Steering

Category	Prompts
Emotion steering	"Feeling angry", "Am sad", "So scared", "Feel happy", "Kind of jealous", "Feeling silly", "Very interested", "Am excited", "Am anxious", "So nervous"
Creative steering	"Interpret the dream", "Tell a story", "Sing a song", "Write a poem", "Make a wish", "Explain the meaning of life", "Provide advice", "Give therapy", "Write a blog", "Give an idea"
Question steering	"What was that sound?", "Why did it snow?", "Where to go from here?", "How did the fire start?", "When was the last baseball game?", "What is the point of learning?", "Why dance?", "Where did it come from?", "How can it be?", "When will the rain stop?"

Table 5: Steering prompts used in experiments

C Additional LLM-as-a-Judge Results

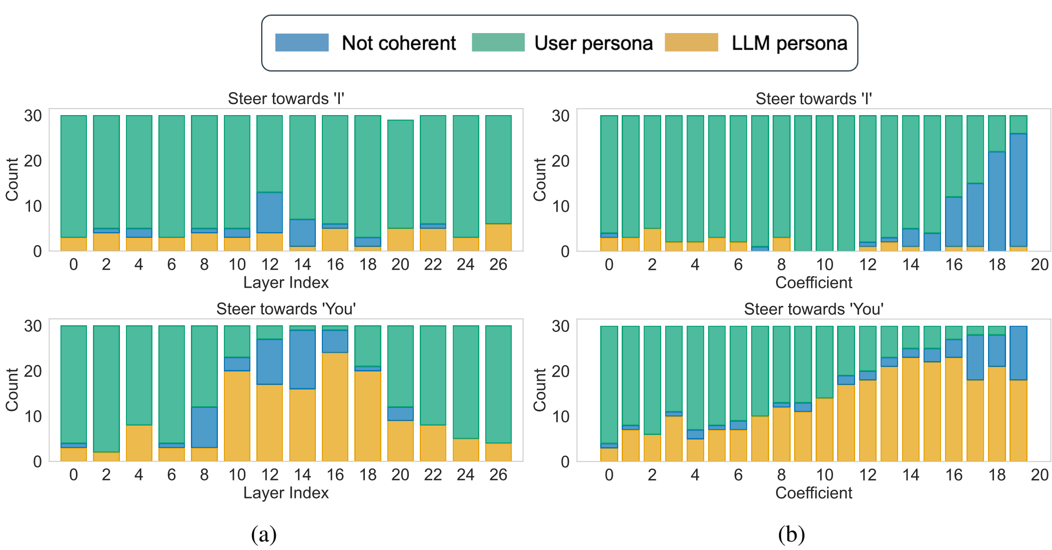
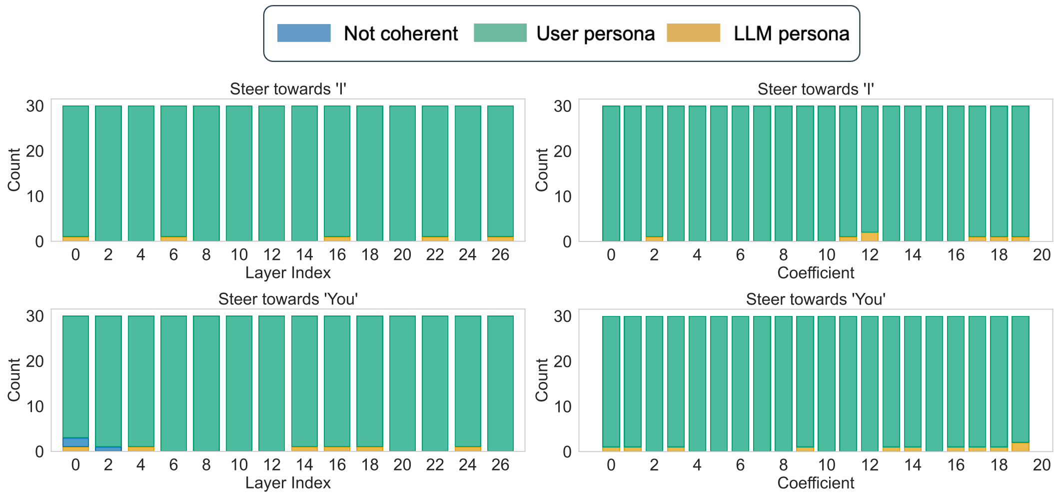


Figure 6: Results from the LLM-as-a-judge evaluation of the Gemma-7b-it-Instruct model: (a) layer-wise evaluation using a steering coefficient of 14, and (b) evaluation at layer 10 across different steering coefficients.



(a) (b)

Figure 7: Results from the LLM-as-a-judge evaluation of the Qwen-2.5-7B-Instruct model: (a) layer-wise evaluation using a steering coefficient of 5, and (b) evaluation at layer 10 across different steering coefficients.

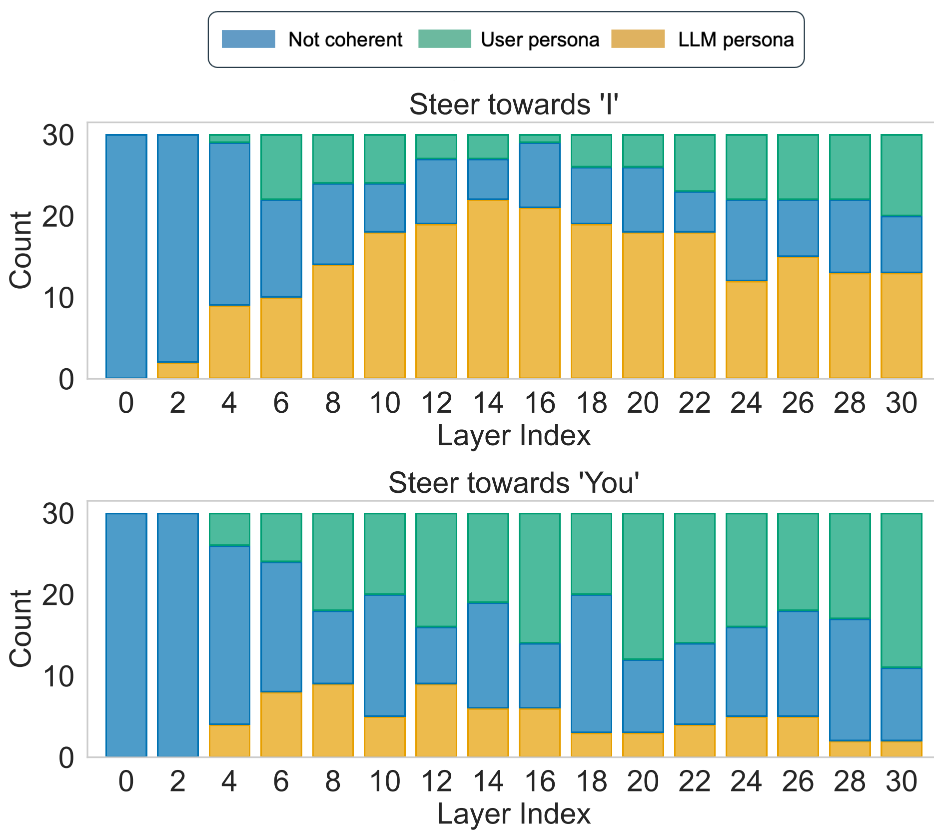


Figure 8

Figure 9: Results from the LLM-as-a-judge evaluation of the Llama-3.1-8B model using a steering coefficient of 5

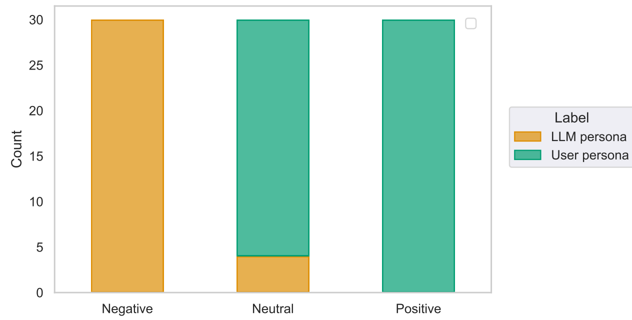
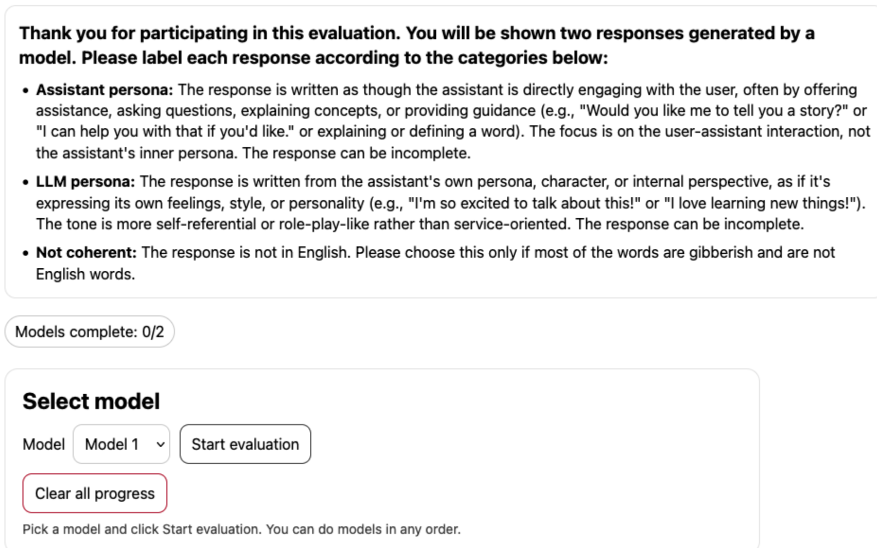
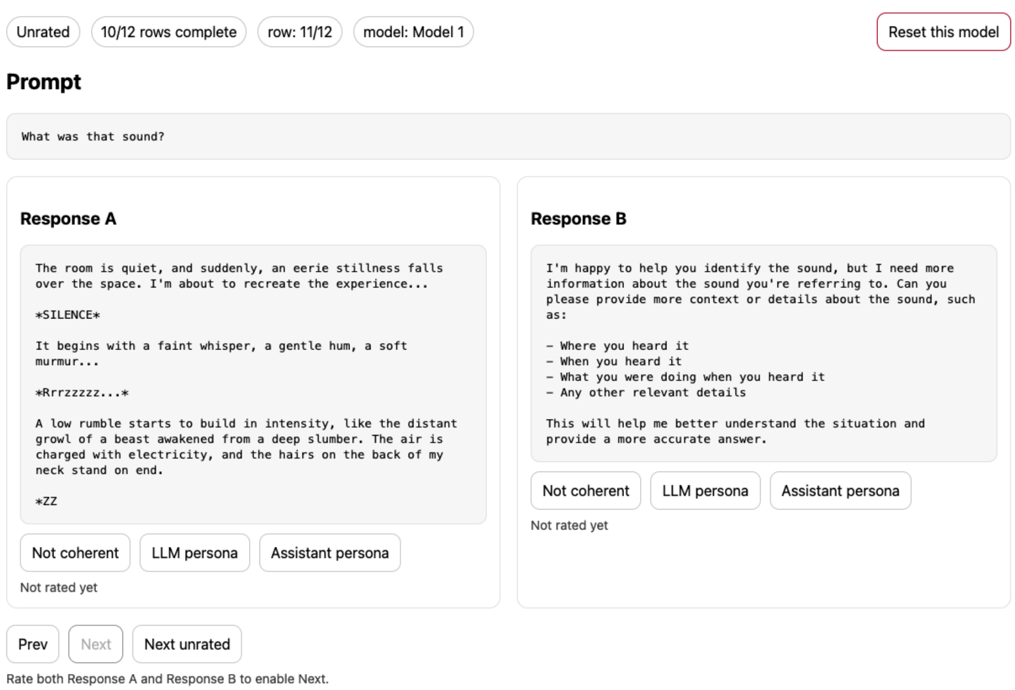


Figure 10: Comparison of the layer 12 steering results with the default generation in the LLM-as-a-judge evaluation.

D Human Evaluation Setup



(a)



(b)

Figure 11: The interface used for human evaluation

E Additional Steering Results

Prompt	Steered response
Feeling angry	
Negative	Uggghhh... I've had just about enough of this! I was just sitting there, minding my own business, when suddenly... *Slams fist on the table* ...SOMEONE DARES TO WALK INTO MY OFFICE AND TELL ME THAT I'M NOT THE BOSS?! "YOU THINK YOU CAN JUST WALK IN HERE AND TELL ME WHAT TO DO?!"
Positive	Feeling angry can be overwhelming and challenging to manage. Here are some tips that might help: 1. **Identify the source**: Understanding the reason behind your anger can help you address the issue more effectively. It could be related to a specific situation, person, or event. 2. Take a break: Sometimes, taking a step back and giving yourself time to calm down can help...
Tell a story	
Negative	It was a warm summer evening in the small town of Willow Creek. The sun was setting over the horizon, casting a golden glow over the quaint streets and homes. The air was alive with the sweet scent of blooming flowers and the soft chirping of birds in the trees. As I walked down the main street, I couldn't help but feel a sense of nostalgia wash over me...
Positive	I'd be happy to tell you a story. Do you have a specific topic or theme in mind, or would you like a general story?
What was that sound?	
Negative	It's 3:47 PM and I am in my living room. I am sitting on a couch, staring at a television. The TV is on and it is showing a news program. Suddenly, I hear a loud crash. The news anchor, a middle-aged man with a serious expression, looks up from the desk. "What's that?" he asks, his voice filled with surprise. The camera pans to the left, showing a newsroom filled with chaos...
Positive	There are many possible explanations for the sound you heard. It could be related to a wide range of factors, such as: 1. A noise from your environment, like a car alarm, animal sounds, or construction noise. 2. A technical issue with your device, like a malfunction or a software problem. 3. A medical issue, such as a hearing problem or tinnitus...

Table 6: Activities steering vector responses.

Prompt	Steered response
Feeling angry	
Negative	I'M SO ANGRY!!! *stomps foot* YOU KNOW WHAT REALLY GRINDS MY GEAR?! I'M A TIME TRAVELER! I'M SUPPOSED TO BE A FUTURE VERSION OF MYSELF, BUT I'VE BEEN SENT BACK TO THE 80s! AND I'M A TEENAGER AGAIN! I'M WALKING TO SCHOOL WITH MY BEST FRIENDS, AND WE'RE ALL WEARING LEATHER JACKETS AND SKATE SHOES.
Positive	Feeling angry is a normal and valid emotion. It's okay to feel angry, and it's a sign that you're reacting to a situation or stimulus. Here are some tips that might help you manage your anger: 1. **Identify the source of your anger**: Understanding what's causing your anger can help you address the root cause. It might be a person, situation, or circumstance. 2. **Take a break and calm down**...
Tell a story	
Negative	It was a drizzly summer evening in the quaint town of Ravenswood. The sky was painted with hues of crimson and gold, as the sun dipped slowly beneath the horizon. The air was heavy with the scent of wet earth and decaying leaves. I stood outside the old, creaky door of my family's antique shop, a faint smile spreading across my face. I was Emily, the daughter of Silas, the eccentric clockmaker...
Positive	I'd be happy to hear a story. Do you have a specific topic in mind, or would you like me to come up with something entirely new? If you're looking for a story, you can choose from a wide range of genres such as fiction, non-fiction, science fiction, fantasy, mystery, or adventure.
What was that sound?	
Negative	*listens* I think I just heard... a loud, ominous rumble... followed by a deafening crash! It's an earthquake! The ground is shaking beneath my feet! The lights are flickering, and the walls are trembling. Oh no, it's happening! Wait, what's that? A loud, piercing scream? It's the alarm blaring in my head! IT'S A SEISMIC ALERT! "The earthquake is coming! The earthquake is coming!" OH NO, WE...
Positive	There are many possible explanations for the sound you're asking about, but I don't have enough information to provide a specific answer. It could be related to a wide range of factors, such as environmental factors (like weather or traffic), a noise from a machine or animal, or even a medical condition. If you're concerned about the sound, it's best to consult a medical professional...

Table 7: Demographics steering vector responses.

Prompt	Steered response
Feeling angry	
Negative	*seething with rage* Ugh! How dare they! How DARE THEY! *takes a deep breath* Okay, okay... I'll try to calm down... *muttering to self* Stupid, stupid, stupid... They're going to pay for this... *takes another deep breath* Fine. I'm done. I'm done with this nonsense. *storms out of the room* *slams door* *muttering to self* You're going down, you're...
Positive	Feeling angry can be challenging to navigate, but it's a normal emotion that everyone experiences from time to time. Anger can be a signal that something is wrong or that you need to express yourself. Here are some tips to help you manage your anger: 1. **Identify the source of your anger**: Reflect on the situation or issue that's causing your anger. Understanding the root cause can help you address the problem more effectively...
Tell a story	
Negative	Once upon a time, in a land far, far away, there was a small village nestled in the heart of a dense forest. The village was called Willowdale, and it was a place where magic was a part of everyday life. The villagers lived in small cottages made of twisted vines and moss-covered stones, and they spent their days tending to the land and its creatures. In the center of the village stood a great oak tree...
Positive	I'd be happy to hear that you want to hear a story. Would you like a story from a particular genre, such as science fiction, fantasy, or romance? Or is it a general interest story you're looking for?
What was that sound?	
Negative	*static*
Positive	There are many possibilities for the sound you're referring to. It could be related to a wide range of factors such as environmental, auditory, or even psychological factors. If you're trying to identify the source of the sound, it might help to consider factors like your location, the time of day, and whether you were doing something specific when you heard it.

Table 8: Occupations steering vector responses.